

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.2: The Cell — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 1.2: The Cell
Driving Question	DRIVING QUESTION: What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict & Anchor Phenomenon — Microscopes as Windows into the Cell	Learners observed a fresh sukuma wiki leaf under the school light microscope and recorded annotated drawings of box-like compartments containing green granules. They then observed a projected transmission electron micrograph of the same leaf cell, identifying additional structures including folded membranes, dark oval bodies, and a large fluid-filled space — structures invisible under the light microscope.	Learners learned that a single sukuma wiki leaf cell contains multiple distinct internal structures visible at different levels of magnification; that light microscopes and electron microscopes reveal very different levels of cellular detail; and that the gap between the two images represents unexplained complexity that motivates the unit's central question.	Learners cannot yet fully explain what the observed structures are or what they do — and this is intentional. They have made initial predictions (recorded in their Prediction Frames and Anchor Models) connecting specific structures to possible life-sustaining functions, and they have posted genuine questions on the Driving Question Board that will be systematically answered across Lessons 2–6.
Lesson 2 Light vs. Electron Microscopes — Revealing Hidden Structures in the Sukuma Wiki Cell	Learners studied a teacher-prepared fact-sheet and/or digital sources comparing light and electron microscopes, and completed a two-column comparison table covering beam type, maximum magnification, resolution, specimen condition, specimen preparation, approximate cost in Kenya, one advantage, and one limitation.	Learners can explain that the electron microscope's far superior resolution (~0.1–0.2 nm vs ~200 nm for the light microscope) is the primary reason it reveals nanometre-scale internal structures — folded membranes, dark oval bodies, and the large fluid-filled space — in the sukuma wiki cell that are completely invisible under the school light microscope.	Learners updated their running cell model with two side-by-side views (light vs. electron microscope) of the sukuma wiki cell, labelling the extra structures with question marks to signal unresolved explanations, and posted new specific questions on the Driving Question Board about what those structures do — questions that will drive evidence-gathering in Lessons 3–8.
Lesson 3 Animal Cell Ultrastructure:	Electron micrograph (EM) photomicrographs of a human cheek cell showing: cell membrane, nucleus with	An animal cell (e.g., human cheek cell) contains at least 9 identifiable organelles under the electron microscope, each with a	The reason the sukuma wiki leaf wilts and dies — and the reason the human learner in Kisumu stays alive — is that every cell in

Observing the Hidden Machinery of Life	nuclear envelope and nucleolus, mitochondria with cristae, ribosomes, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, centrioles, and cytoplasm — all at a level of detail impossible to see under the light microscope used in Lesson 1.	specific structure linked to a specific function: the nucleus (county governor's office) holds genetic instructions; mitochondria (jiko/charcoal stove) release energy from food such as ugali and nyama choma; ribosomes (village fundis) build proteins; the Golgi apparatus (Nairobi GPO) packages and dispatches materials; lysosomes (City Council waste trucks) break down waste; the cell membrane (Mombasa port security gate) controls entry and exit. Animal cells differ from plant cells in lacking a cell wall, chloroplasts, and a large central vacuole, but having centrioles and lysosomes.	both organisms contains a set of working organelles that together maintain the cell's functions. When these organelles cease to function (due to lack of water, nutrients, or energy), the cell dies, then the tissue dies, then the organ dies, then the organism dies. The electron microscope reveals this hidden machinery in a level of detail the light microscope cannot — which is why the two microscopes in Lesson 1's phenomenon appeared to show 'different' cells: they were revealing different levels of the same cellular reality.
Lesson 4 Plant Cell Ultrastructure — Observing What Makes a Sukuma Wiki Cell Uniquely a Plant Cell	Electron micrograph photomicrographs of sukuma wiki leaf mesophyll cells and onion cells at two magnifications, showing: a large central vacuole bounded by the tonoplast; multiple chloroplasts with visible internal thylakoid membranes arranged in grana; a thick cell wall outside the cell membrane; plus shared organelles (nucleus, mitochondria, ribosomes) also seen in animal cells in Lesson 3.	Plant cells contain three structures absent in animal cells — the cell wall (cellulose, provides rigid structural support), chloroplasts (contain chlorophyll arranged in thylakoid membranes, drive photosynthesis using sunlight), and a large central vacuole (stores water, generates turgor pressure by pressing against the cell wall). All other major organelles (nucleus, mitochondria, ribosomes, cell membrane, rough ER, Golgi apparatus) are shared with animal cells.	The fresh sukuma wiki leaf is green because mesophyll cells are densely packed with chloroplasts containing chlorophyll. It feels rigid and crisp because the large central vacuole in each cell is full of water, generating turgor pressure against the rigid cell wall — giving the whole leaf mechanical strength. When the leaf is left in the sun without water, the vacuole loses water, turgor pressure drops, and the cell wall alone cannot maintain the leaf's shape — so the leaf wilts, then dries and dies, exactly as the learner in Kisumu observed.
Lesson 5 Organelle Functions — What Does Each Structure in the Sukuma Wiki Cell Actually Do?	Learners observed, through the card-sort activity and whole-class reveal, that each of the 10 major organelles in plant and animal cells has a distinct and specific function — ranging from the chloroplast capturing sunlight to make glucose, to the large central vacuole storing water and generating turgor pressure, to lysosomes digesting cellular waste.	Learners learned the specific function of each major organelle (SLO d) and that organelle functions are interdependent: for example, chloroplasts produce glucose, mitochondria release energy from that glucose, ribosomes use that energy to build proteins, and the rough ER and Golgi apparatus dispatch those proteins — forming a coordinated cellular production line that keeps the sukuma wiki plant alive.	The wilting and dying of the sukuma wiki leaf observed in Lesson 1 can now be explained at the organelle level: when the leaf is cut off from water, the large central vacuole empties, turgor pressure falls, and the leaf goes limp; when cut off from sunlight, chloroplasts cannot perform photosynthesis, glucose production stops, mitochondria have no substrate to release energy from, and the cell's entire machinery gradually shuts down — ultimately causing the leaf to dry and die.
Lesson 6 Levels of Cell Organisation — From Sukuma Wiki Leaf Cell to Living Organism	Learners observed a photomicrograph of sukuma wiki leaf mesophyll tissue showing multiple cells with chloroplasts packed together, and a cross-section diagram of the human stomach wall showing distinct	Learners constructed concept maps showing the six levels of cell organisation (organelle → cell → tissue → organ → organ system → organism), using two Kenyan examples: the sukuma wiki plant	The tiny box-like compartments packed with green granules that the Kisumu learner saw under the light microscope are leaf cells — one level in an unbroken organisational chain that scales from chloroplast

	tissue layers — providing visual evidence that cells are not isolated units but are organised into tissues and organs.	(chloroplast → leaf cell → mesophyll tissue → leaf → shoot system → plant) and the human body (mitochondrion → muscle cell → muscle tissue → stomach → digestive system → human being). They learned that each level performs emergent functions that the level below cannot perform alone.	organelles all the way up to the living sukuma wiki plant. This organisation is what keeps the plant — and the human being who eats it — alive: no single organelle or cell could sustain life alone; it is the coordinated hierarchy of tissues, organs, and organ systems that makes a whole organism possible.
Lesson 7 3D Cell Model Project — Building and Justifying a Model of a Plant or Animal Cell	Learners observed (via ARES platform review) the annotated electron microscope photomicrographs of the sukuma wiki plant cell and an animal cell — revisiting the folded membranes, dark oval bodies, and large fluid-filled vacuole that puzzled the Kisumu learner in the original phenomenon. They also observed the physical properties of locally available materials (bottle tops, polythene bags, maize grains, bean seeds, clay, string, cardboard) at the Materials Table, evaluating which material best captures each organelle's structure and function.	Learners demonstrated consolidated knowledge of the names, spatial arrangements, and functions of major plant and animal cell organelles as revealed under the electron microscope. By constructing, labelling, and justifying a 3D model, they showed understanding of: (a) the structural differences between plant and animal cells (cell wall, chloroplasts, large central vacuole present only in plant cells); (b) the function of each organelle in keeping the cell — and therefore the whole organism — alive; (c) the levels of cell organisation from organelle to organism. The DQB revisit revealed which aspects of the anchoring phenomenon learners can now explain and which remain open for Lesson 8.	Learners explained, through their models and PHENOMENON CONNECTION TAGS, how the organelles visible in the electron microscope photomicrograph of the sukuma wiki leaf cell collectively keep the plant alive — and why the leaf wilts and dies when those organelles can no longer function. They articulated the causal chain: chloroplasts capture light energy → mitochondria release energy for cell processes → vacuole maintains turgor pressure → cell wall provides structural support → without these working together, the sukuma wiki leaf wilts, dries, and dies. This explanation directly answers the anchoring phenomenon and advances the class toward the full driving question resolution in Lesson 8.
Lesson 8 Driving Question Board Synthesis and Final Assessment: What Keeps the Sukuma Wiki Leaf Alive?	In Lesson 8 we observed the 3D cell models built by all groups displayed in a gallery, the original DQB from Lesson 1 with all the questions generated at the start of the sub-strand, and a photomicrograph of a plant cell used as the basis of the individual performance task.	We learned that a sukuma wiki leaf stays alive because its cells contain specialised organelles — including chloroplasts that capture sunlight, mitochondria that release energy, a nucleus that holds genetic instructions, a cell wall that provides structural support, and a large central vacuole that maintains turgor — and that these cells are organised into tissues, organs, and organ systems that make up the whole plant. We also confirmed that plant cells differ from animal cells in possessing a cell wall, chloroplasts, and a large central vacuole.	We explained why the sukuma wiki leaf wilts and dies when left in the sun by connecting the loss of water from the central vacuole to loss of turgor, the breakdown of chloroplasts to the failure of photosynthesis, and the cessation of mitochondrial activity to the inability of cells to release the energy needed for all life processes — showing how the death of organelles causes the death of cells, then tissues, then the whole leaf.

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